Assessment of UV/VIS-spectrometry performance in combined sewer monitoring under wet weather conditions

Gamerith V.[1*], Steger B.[1], Hochedlinger M.[2] and Gruber G.[1]

[1]*Graz University of Technology, Institute of Urban Water Management and Landscape Water Engineering, Stremayrgasse 10/1, A-8010 Graz, Austria*
[2]*Linz AG – Wastewater, Wiener Straße 151, A-4021 Linz and Urban Technologies, University of Applied Sciences FH Joanneum, Werk-VI-Straße 46, A-8605 Kapfenberg, Austria*
**Corresponding author, e-mail gamerith@sww.tugraz.at*

ABSTRACT
Ultraviolet-visible (UV/VIS) spectrometry allows deriving continuous high resolution data in-situ on several waste water compounds. However, a correlation between absorption and the target parameter has to be defined and especially the site and event-specific *wastewater matrix* impacts on the derived parameters. The aim of this study was to estimate the reliability of in-situ measurements in combined sewers during highly dynamic wet-weather conditions where important variations in concentrations as well as in the wastewater matrix are encountered. Pollutant concentrations for chemical oxygen demand and total suspended solids were assessed by i) an in-situ UV/VIS spectrometer and ii) manual sampling and lab analysis. Based on this analysis uncertainties when using a predefined, *global calibration* were estimated and the effect of local calibration by simple regression and coupling with a global optimiser was assessed. It was shown that the *global calibration* provided by the manufacturer lead to a percentage error of up to 50% compared to conventionally obtained composite samples. With local calibration on single events, these events could be fitted satisfactorily. Validation on other events, however, resulted in higher errors than using the global calibration. Errors obtained with a calibration on all available data points were in the order of magnitude of 25% to 30%.

KEYWORDS
online monitoring; pollutant concentrations; sensor calibration, urban wastewater; UV/VIS spectrometry; wet weather flow.

INTRODUCTION
Assessing pollutant concentrations in combined sewers by ultraviolet-visible (UV/VIS) spectrometry shows a number of advantages compared to traditional sampling methods. Continuous high resolution data on several waste water compounds can be derived at relatively low costs compared with automatic sampling and lab analysis. However, a correlation between absorption and the target parameter has to be defined. Especially the site specific wastewater composition (*wastewater matrix*) impacts on the derived parameters. Several studies discuss the issue of probe calibration and highlight the importance of adaption to the local wastewater matrix, e.g. Gruber *et al.* (2006), Rieger *et al.* (2006) or Torres and Bertrand-Krajewski (2008). Winkler *et al.* (2008) discuss the uncertainties in UV/VIS measurements based on lab measurements for raw wastewater samples and conclude that the device is comparable robust if the matrixspecific relationship between measured absorption and target parameter concentration is determined by a suitable correlation model.
The main goal of this study was to estimate the reliability of in-situ measurements in combined sewers during highly dynamic wet-weather conditions, where important variations of measured pollutant concentrations as well as event-specific changes in the wastewater matrix are encountered. To address this question water quality was analysed by i) an in-situ UV/VIS spectrometer installed directly in the sewer system and ii) manual sampling with a peristaltic-

operated sampler at the same location and corresponding time stamps with subsequent standardised lab analysis. All samples were analysed on chemical oxygen demand (COD) and total suspended solids (TSS).
This paper focuses on the discussion of the global probe calibration provided by the manufacturer and the effect of local calibration when using samples from single storm events for calibration: It is shown that for the study data global calibration lead to a percentage error of up to 50% with systematic behaviour compared to conventionally obtained composite samples. With local calibration, single events could be well fit. Validation on other events, however, showed significantly higher errors than using the global calibration. Using all available data lead to better results than the global calibration. However, as validation of this calibration is still lacking the results have to be interpreted with caution. For operation, caution is advised when calibrating the probe locally to samples from one single rainfall event, as not all possible matrix effects can be assessed. This might lead to higher overall errors than using the provided global calibration. Especially for future applications as e.g. real-time control based on pollution fluxes or concentrations, information on the reliability and accuracy of such measurements is crucial.

METHODS

Sensor and Measurement Setup

All samples were taken at the *Graz Sewer R05* measurement station located at a combined sewer overflow (CSO) at the outlet of the urban catchment *Graz West R05* in Graz, Austria. The catchment with its 19 500 inhabitants is drained by a combined system and covers approximately 4.6 km² wherein 1.3 km² are impervious. The catchment and the measurement station are exhaustively described e.g. in Gruber *et al.* (2005) and Gamerith *et al.* (2011). The UV/VIS probe used in this study is a spectro::lysers from the company s::can. Based on the measured absorptions in different wavelength ranges, so called equivalent concentrations can be calculated for i) organic matter as e.g. COD_{eq} and total organic carbon, ii) TSS_{eq} and iii) $NO_{x,eq}$. A detailed description of the sensor can be found e.g. in Langergraber *et al.* (2003). The target parameter concentration is calculated by Equation 1. Different subsets of wavelengths *i* are used for the different compounds.

$$C_{eq} = \sum_{i=1}^{n} w_i \cdot A_i + K \qquad \textbf{Equation 1}$$

with C_{eq} … equivalent concentration (mg/L); w_i… factor for wavelength *i;*
A_i … measured absorption for wavelength *i* (1/m) and *K* … offset

The in-situ spectrometer is installed in a floating pontoon, directly in the sewer at a CSO. Reference samples were collected close to the UV/VIS probe by an automatic peristaltic-operated sampler. One sample corresponds to a composite sample over the filling time period of one sampler bottle which takes between of 3 and 5 minutes. Over the sampling duration, at least three absorption spectra were recorded by the in-situ UV/VIS spectrometer. All recorded spectra were analysed manually on erroneous recordings and then averaged for calculating the derived water quality parameters.
The manually taken samples were analysed in the laboratory on COD and TSS as soon as possible after the site sampling to reduce and prevent changes in the wastewater matrix. Therefore the samples were cooled and transported about 1.5 km to the lab and homogenized in the lab with an *Ultraturrax*. COD was evaluated with a *Lange COD cuvette test*. TSS were determined by filtering of 50 ml homogenized samples at 8 bar, subsequent drying at 105° C and weighing. All samples were tripled to estimate the lab analysis uncertainty.
In total 36 samples from 6 storm events as shown in Table 1 were obtained and used for probe calibration. Measured concentration for TSS range from about 60 to 730 mg/L, with a median of 153 mg/L. COD concentrations are wider spread, ranging from about 150 to 1400 mg/L, with a median value of 310 mg/L.

Table 1: Events and number of samples for UV/VIS probe calibration

event ID	sampled IDs	ID range	date	first sample	last sample
-	#	-	yyyy-mm-dd	hh:mm	hh:mm
1	5	1 to 5	2008-09-25	14:33	15:00
2	8	6 to 13	2008-09-25	18:48	19:52
3	3	14 to 16	2008-11-13	15:54	16:50
4	8	17 to 24	2008-11-21	18:28	19:39
5	8	25 to 32	2008-12-01	11:39	12:30
6	4	33 to 36	2009-03-06	10:07	11:06

Probe calibration procedure

Local probe calibration was carried out after validation of the recorded raw spectra by i) simple regression between mean lab concentration values and UV/VIS values obtained from a *global calibration* provided by the manufacturer and ii) identification of the factors w_i and offset K given in Equation 1 by an optimiser suited for automated calibration.

The application of regression methods is a practical approach that can also be applied easily by end-users who do not have access to the measured absorption spectra. A linear and a non-linear regression were performed in the software [R].

The optimiser was used for local probe calibration in order to compare calibration on single events, using different sets of wavelengths etc. It is a global optimiser based on evolutionary strategies and allows multi-objective optimisation. It was developed by Muschalla (2006) and has so far been successfully applied in calibration studies e.g. by Muschalla (2008) and Gamerith *et al.* (accepted). It is part of the optimisation framework BlueM.OPT (Bach *et al.*, 2009) developed at TU Darmstadt. A class for UV/VIS probe calibration was implemented in the BlueM.OPT framework recently (Gamerith, 2011).

One can argue that regression methods as e.g. discussed in Bertrand-Krajewski *et al.* (2000) are better apt for probe calibration than global optimisation algorithms. An intensive discussion on the application of regression methods in UV/VIS probe calibration is given in Hochedlinger (2005). The optimiser was chosen mostly due to its flexibility as i) it allows minimising several objective functions, ii) it is flexible to run for different sets of parameters (e.g. wavelengths, factors, offset) and iii) allows simple grouping of parameters. Provided a sufficient number of iterations, the algorithm showed to perform equally well to regression methods. In this study, different sets of wavelengths i and weighing factors w_i were used in local probe calibration. Calibration was done separately for each event and against all the available data to evaluate the impact of the calibration data set on the probe calibration uncertainties. In total, 32 optimisation runs were performed where the following combinations were investigated:

- Variation of the wavelength ranges: i) Wavelength ranges as proposed by the manufacturer for COD and TSS and ii) the whole range of available wavelengths.
- Variation of the weighing factors w_i: i) grouped weighing factor, constant over all considered wavelengths and ii) variable weighing factors for each considered wavelength. In addition different parameter ranges for w_i were tested.

In the following, only results for grouped factors are presented, as variable factors can lead to over-parameterisation. The used ranges for the wavelengths and the factor values are not stated here as they are classified information from the manufacturer of the probe. Details on the optimisation runs are given in Steger (in preparation). Due to limited space and the higher errors in lab analysis for TSS, in this contribution results are presented for COD only. The general behaviour discussed below, however, is valid for both COD and TSS.

RESULTS AND DISCUSSION

UV/VIS spectra analysis and validation

As already identified in Hochedlinger (2005), validation of the raw spectra is crucial in order to obtain viable results. As stated above, one sample ID from the UV/VIS probe is composed of three recorded spectra that are then averaged. Figure 1 shows an example for the impact of an erroneous measurement. It shows three recorded absorption spectra and the calculated averaged spectrum for ID3. The 2nd measured absorption spectrum can be identified as erroneous, leading to strong influence on the averaged spectrum that is then used for calculating the derived concentration value. All 36 IDs were analysed and all erroneous spectra were removed manually.

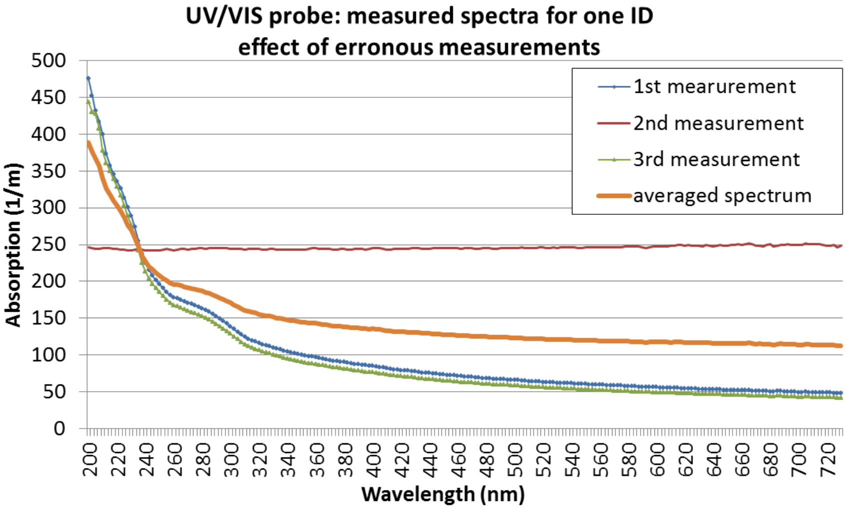

Figure 1: UV/VIS probe - impact of erroneous measurement on averaged spectrum

Lab analysis uncertainties:

For an estimation of the uncertainties in the lab analysis, a triple analysis was performed for each sample. From each triple, mean and standard deviation were calculated. Figure 2 shows a boxplot of the percentage differences between the mean and the 0.95 quantile for the 36 samples for TSS and COD. In order to increase legibility, different scales were chosen for TSS and COD. It can be seen that the variability in the lab analysis is significantly lower for COD than for TSS. For COD the 0.95 quantile has a median difference from the lab mean value of about 5%, the highest value for one single sample is at 21%. Deviations for TSS are about twice as high, with a median difference of 10%. For three of the samples this difference is in a range of 60%.

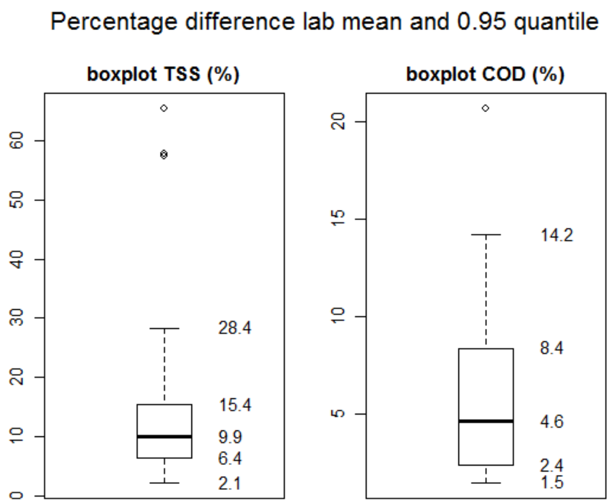

Figure 2: Boxplots of percentage difference between mean and 0.95 quantiles for the lab analysis of TSS and COD for the 36 samples

Evaluation of the global probe calibration

For evaluation of the global probe calibration, COD_{eq} concentrations were calculated from the recorded spectra by using the so called global calibration *INFLUENTV120* provided by the manufacturer. In this context, a defined range of wavelengths from the recorded spectrum as well as the weighing factors and offset as described in Equation 1 is understood by *global calibration*. The equivalent concentrations are then calculated based on this global probe calibration.

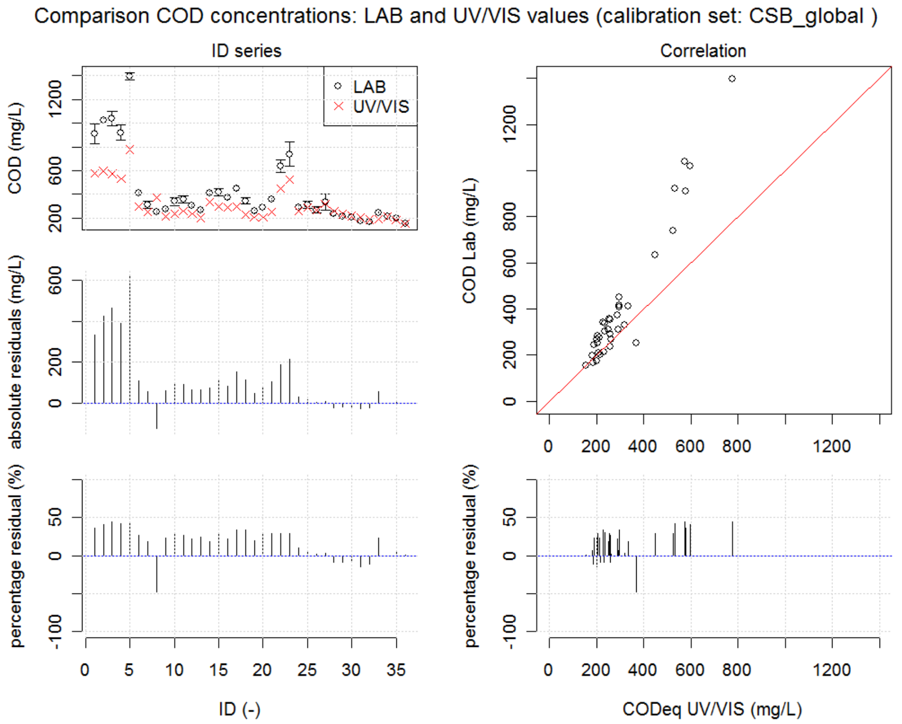

Figure 3: Evaluation of global UV/VIS probe calibration – COD concentrations

Figure 3 show the results obtained with the global probe calibration for COD. The left hand side of the plots shows the measured concentrations from the UV/VIS probe and the corresponding lab values (with whiskers for the 0.05 and 0.95 quantiles), the absolute residuals and the percentage error for each of the 36 samples. The right hand side shows a correlation plot of the UV/VIS and mean lab concentration values and the sorted resulting percentage residuals. The evaluation shows that with the available data the global calibration led to a percentage error of up to 50% compared to the lab values. The error has systematic behaviour as is visible from the correlation and residual plots. With this calibration, lower concentrations are better fit (i.e. sample IDs 24 to 36), where the UV/VIS values lie with one exception within the 95% confidence interval of the lab analysis. UV/VIS equivalent concentrations are generally underestimated with this calibration.

Local probe calibration: simple regression methods

A linear and non-linear regression between lab mean concentrations and the concentrations obtained from global calibration were performed. Values from the global calibration were corrected by the obtained regression coefficients and the results for the two regression methods compared. Table 2 shows the results for the regression coefficients and the resulting relative residual errors (0.05 and 0.95 quantiles) with the corrected UV/VIS data. While both regressions lead to a significant reduction of the errors in the investigated concentration range, linear regression is not suited for extrapolation as applying it to low concentration values would result in negative values due to the negative intercept (*a* – coefficient). The power regression, on the other hand might influence extrapolation to higher values. This, however could only be addressed in detail with additional data covering the lower and higher concentration ranges.

Table 2: Regression for lab measurements and UV/VIS global calibration values

Regression	Formula	Obtained regression coefficients		relative residual errors (lab – corrected UV/VIS)	
		a	b	0.05 quantile	0.95 quantile
Linear	$y = a + b \cdot x$	-180.93	1.96	-29.7%	24.8%
Non-linear power function	$y = a \cdot x^b$	0.17	1.36	-27.9%	17.8%

Figure 4 shows a plot of the obtained corrected COD_{eq} UV/VIS values with non-linear regression from global calibration. Compared to the global calibration, relative residual errors are reduced to an order of magnitude of 25% to 30% and the error becomes less systematic.

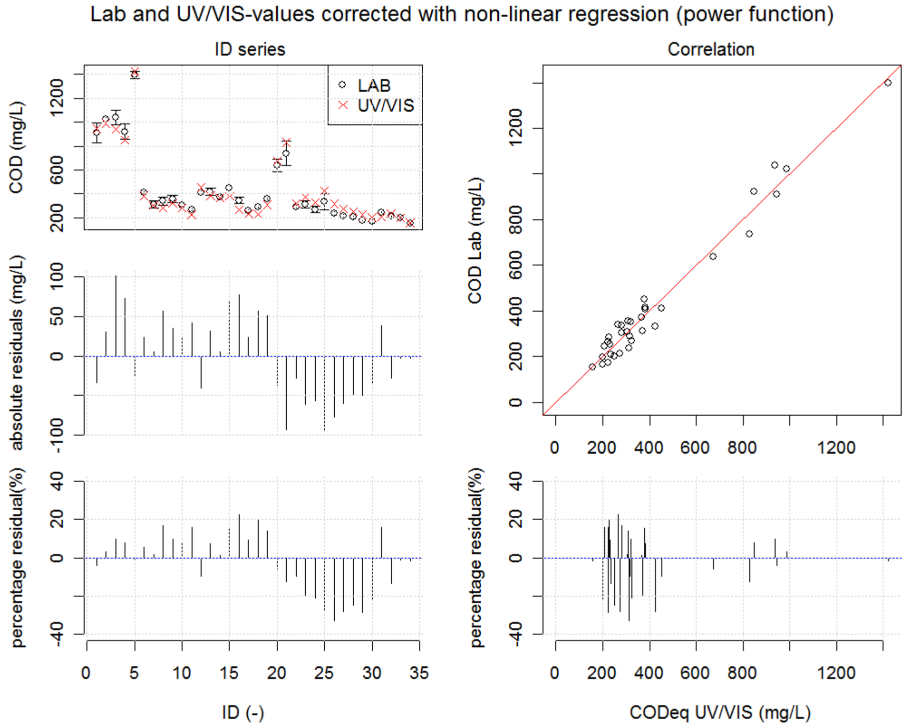

Figure 4: Evaluation of global UV/VIS probe calibration – COD_{eq} values corrected by non-linear regression

Local probe calibration: coupling with an optimisation algorithm

Figure 5 shows the results for calibration when using sample IDs from one single storm event (IDs 17 to 24) using all available wavelengths and a grouped weighing factors, constant for all considered wavelengths. The vertical orange lines in the upper left plot delimit the IDs used for calibration. The corresponding values in the scatter correlation plot are also shown in orange.

With this calibration, the calibrated IDs can be fitted satisfactorily with percentage errors < 10% within the error estimates from the lab analysis. In addition events 2 and 3 where a similar concentration range was measured are also well fit. However, validation on events 1and 5 shows significantly higher errors than using the global calibration with up to 100% percentage error. This behaviour was also observed when calibrating to the other events separately.

Figure 6 shows the results obtained when using all sampled IDs in a calibration with all available wavelengths. Here, the overall fit is generally better than when using the global calibration. Errors are in the same range as for the values corrected by non-linear regression. The behaviour shows a systematic error where higher values are slightly underestimated and lower values overestimated by the UV/VIS probe. While the calibration shows significant improvement compared to calibration on single events it has not yet been validated against additional sample data to prove its validity.

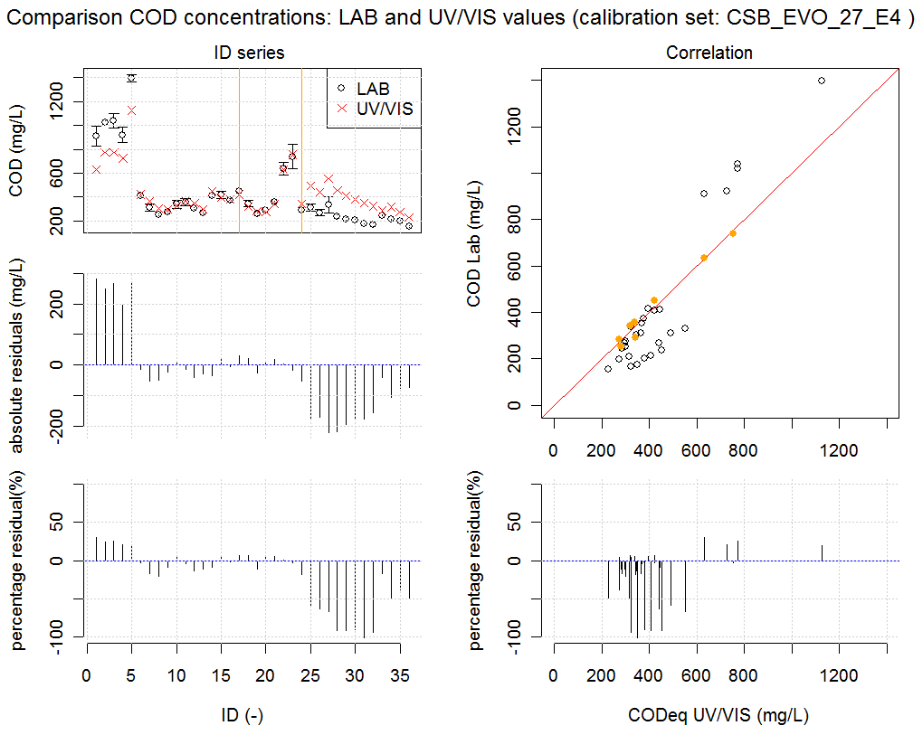

Figure 5: UV/VIS COD calibration for event 4 using all available wavelengths and a grouped weighing factor.

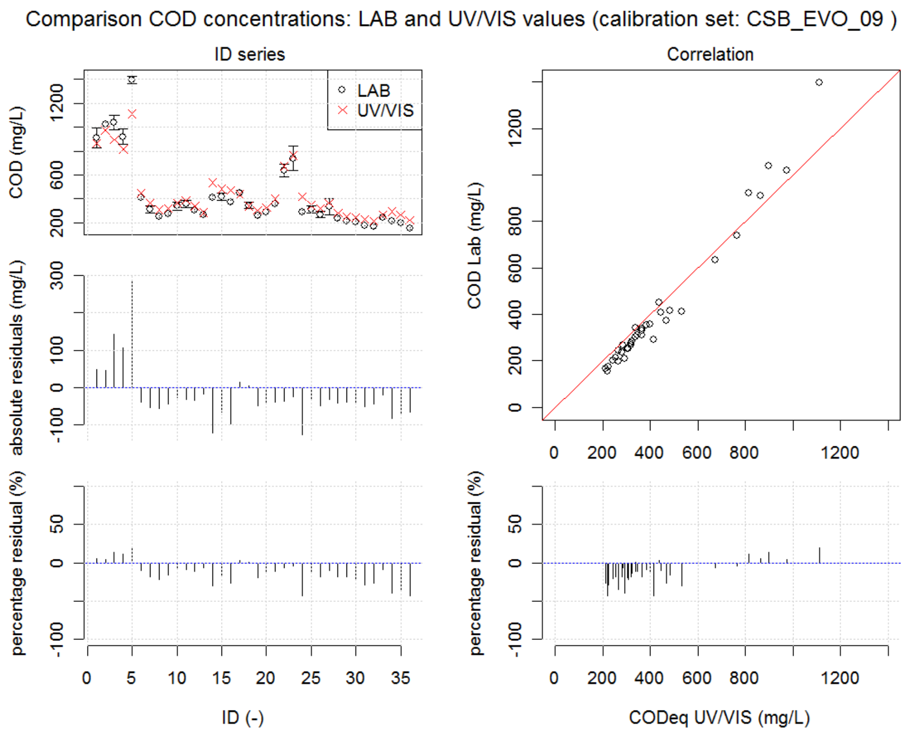

Figure 6: UV/VIS COD calibration using all sampled IDs and all available wavelengths

CONCLUSION

The evaluation of the global calibration highlights the importance of local probe calibration as significant errors of up to 50% for COD with systematic behaviour were identified when using the global probe calibration provided by the manufacturer. With simple regression between lab concentration values and concentration values obtained by the global probe calibration, the errors could be reduced to an order of magnitude of 25 to 30%. However, special attention has to be paid when extrapolating concentration ranges from the regression results: i.e. in the presented example correction by linear regression would lead to negative values for low concentration values. Several optimisation runs were performed with the BlueM.OPT optimiser for local probe calibration in order to evaluate the calibration on single events, using different wavelength ranges and weighing factors. The following major finding can be stated:

- Caution is advised when calibrating UV/VIS probes locally to samples from one single rainfall event or events with similar concentration ranges. In that case not all possible effects and variations in the wastewater matrix during wet weather conditions can be assessed. For the presented results the calibration events were well fitted (percentage errors < 10%). However, errors for the validation samples reached over 100% with systematic behaviour. Hence, higher total errors are obtained than when using the global calibration.
- As already highlighted in previous studies, the validation of the raw spectra is crucial to avoid errors in the averaged spectrum used for calculation of the concentration values.
- For single events, no difference in calibration quality was identified when using more wavelengths then the ones proposed by the manufacturer.
- For the available data no significant amelioration of the results compared to a correction by simple regression could be identified.

Future work should focus on the collection of additional samples to validate the calibration especially in concentration ranges that are not yet covered by the samples. In addition it is advised to implement automated validation of the raw spectra in the data validation process.

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